

# Display Data: Histograms

Lesson 8-6

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_ **Class:** \_\_\_\_\_

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## TODAY'S OBJECTIVES

**Content Objective:** I can make and read a histogram to display data in intervals.

**Language Objective:** I can explain my histogram using the words histogram, frequency, interval, and distribution.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Histogram

Frequency

Interval

Distribution

Variability



Tap any word to see what it means and a picture.

1

A bar graph that shows how often data falls into equal intervals is a

2

How many data values fall in an interval is the

3

An equal-sized range of values used to group data is an

4

The overall pattern of how data is spread out is the

5

How much the data values differ is the

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**As you build the frequency table with intervals of 10, what does the height of each bar in the histogram represent?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Histogram** — A bar graph that groups data into equal ranges. The bars touch.

*Histograma — Una gráfica de barras que agrupa datos en rangos iguales. Las barras se tocan.*

- ☐ **Frequency** — How many times a value shows up.

*Frecuencia — Cuántas veces aparece un valor.*

- ☐ **Interval** — A range of numbers used to group data.

*Intervalo — Un rango de números para agrupar datos.*

- ☐ **Distribution** — How the data is spread out.

*Distribución — Cómo están repartidos los datos.*

- ☐ **Variability** — How spread out the numbers are.

*Variabilidad — Qué tan separados están los números.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The height of each bar shows the \_\_\_\_ of that interval.**

*La altura de cada barra muestra la \_\_\_\_ de ese intervalo.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** A histogram groups data into equal intervals so large data sets become readable.

**Evidence:** Students explain grouping into intervals makes 30 values readable and that each bar's height is the frequency for that interval.

**Reasoning:** This evidence connects the definition of histogram to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The height of each bar shows the \_\_\_\_ of that interval.**

*La altura de cada barra muestra la \_\_\_\_ de ese intervalo.*

- **The tallest bar is the interval where \_\_\_\_.**

*La barra más alta es el intervalo donde \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*